



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-300-1

Date: 19-7-2022

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SPECIFICATION

FOR

MEDIUM/LOW VOLTAGE LIVE-WORKING DETECTOR

RANGE FROM 380 V TO 36 kV

Issue: Jul -2022/ Rev- 1



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1. SCOPE:

This specification covers the minimum technical requirements for design, engineering, manufacturing and testing (low/medium) voltage detector that give clear continuous indication of the state "voltage presence" or "voltage absence" by means of the change of the status of signal. The indication shall be visual and audible. The voltage detector should be designed to detect AC voltage in range from (380 V to 36 kV) for indoor and outdoor installations and manufactured in such away so as to avoid any danger to the operating personnel during use and under normal conditions.

2. APPLICABLE STANDARDS:

Unless otherwise specified in this specification, voltage detector should comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the applicable IEC standards as following

Standard No.	Description
IEC 61243-1	It is applicable to portable voltage detector with or without built-in power sources to be used on electrical systems for voltage of 1K.V. to 765 K.V. A.C. and frequencies of 50 HZ and/or 60HZ. It applies only to voltage detectors of capacitive type used in contact with the part to be tested.
IEC 61235	It is applicable to insulated stick

3. ENVIRONMENTAL CONDITIONS:

The performance of voltage detector should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Minimum ambient temperature	-5°C
Maximum ambient temperature	50°C
Maximum relative humidity	95%.



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4. CONSTRUCTION REQUIREMENTS:

1. The voltage detector as complete device shall include at least the following elements: indicator, contact electrode, limit mark, insulating element, hand guard and handle
2. Nominal voltage range: 380V TO 36K.V.
3. The threshold voltage (U_t) shall satisfy the relationship:
($0.10U_n \text{ max.} < U_t \leq 0.45U_n \text{ min.}$), and must be less than or equal to 50V ($U_t \leq 50\text{v.}$)
4. Nominal frequency: 50HZ $\pm$ 3%
5. Response time less than 1 sec.
6. Time rating $\geq$ 5min.
7. Grip force $\leq$ 200 n.
8. Deflection $\leq$ 10% of detector length.
9. Non – response to a DC voltage.
10. Leakage current $\leq$ 0.5mA.
11. Capacitive type used in contact with the part to be tested.
12. The voltage detector shall give the visual indication with high intensity led's with flashing arrangement, thereby providing the suitable indication in day light.
 - The intensity of the light shall be 50000 lux $\pm$ 10% for outdoor type voltage detector.
 - The intensity of the light shall be 1000 lux $\pm$ 10% for indoor type voltage detector.
13. The voltage detector shall give audible sound for the presence of live line so that it can be suitably used in the populated and noisy areas for easy operation.
 - The sound pressure level shall be $>80\text{dB}$
14. The indication of the detectors shall be with one active signal which gives an indication of the condition “voltage present “ and shall have stand-by state .
15. The voltage detectors should intelligently sense the voltage at overhead line/switch yard without any adjustable pot setting.
16. A self-test button to be provided to assess healthiness of voltage detector.
17. The voltage detector shall be capable of performing complete self-diagnostic check to ensure the working of the probe before taking on site. The self-diagnostic feature will enhance safety at hazardous areas.
18. The contact electrode readily may be interchangeable or completed with contact electrode extension in order to adapt the voltage detector to different uses.
19. The insulating material used in manufacturing of voltage detector should be non-hygroscopic, non-ageing and of tested quality.
20. The voltage detectors must provide protection against bridging.
21. The detector shall be constructed so that the indicator cannot be damaged or shut off as result of spark discharge.
22. The voltage detector shall not indicate “voltage present” for usual values of interference voltages.



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23. The voltage detector probe should have universal adaptor with suitable length for attachment of the insulating stick.
24. Indoor and outdoor use by appropriate insulating stick to the detector voltage range. Connected to the stick by universal connecting adaptor
25. The insulating stick should be of premium quality insulated material
26. The total length of the insulating stick must be $\geq 1250\text{mm}$
27. The insulating stick shall be shortened for transport by telescoping or by disassembly.
28. The outer diameter of the insulating stick should be $32\text{ mm} \pm 1.5\text{ mm}$.
29. The voltage detector with built-in power source shall give clear indication until the source is exhausted, unless its usage is limited by an indication of non-readiness or automatic shut-off
30. Dry battery shall be used. It shall have suitable voltage ratings and shall be commercially available at market and easily replaceable at field level.

5. MARKING:

The marking should be durable and easily legible. Compliance is checked by inspection, The following information should be marked on all voltage detector name plate

- Range of rated voltage
- Nominal frequencies
- Indication of category (S or L)
- Indication of type “indoor” or “outdoor”
- Manufacture’s name and trade mark
- Serial no.
- Climate class – normal out door
- Self-test guide lines
- Year of production

6. TESTS:

All test results shall be provided for review and acceptance by ...EDC.

6.1 Type Tests:

Voltage detector; according to IEC 61243-1

- Measurement of leakage current before humidity test.
- Humidity test.
- Measurement of leakage current after humidity test.
- Dry and wet power frequency voltage withstand test
- Insulation resistance test
- High temperature (dry heat) test
- Low temperature (cold) test



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Telescopic stick; according to IEC 61235

- DRY/WET power frequency voltage withstand test
- Crushing test
- Bending test
- Dye penetration test.

6.2 Routine Tests

Voltage detector; according to IEC 61243-1

- External visual inspection.
- Dielectric test

6.3 Sampling Tests

The test shall be performed on three complete voltage detectors. If more than one voltage detector does not pass, the test fails. If only one voltage detector fails the entire sequence for the test shall be repeated on three other voltage detectors. If again any of the voltage detectors does not pass the test is considered to have failed.

7. DRAWING AND CATALOGUES:

- All necessary drawings including layout, dimensions should be submitted with the offer.
- Technical catalogues containing the technical data should be submitted.
- Installation and maintenance manual at delivery.

8. GUARANTEE

- The supplier should guarantee the voltage detectors against all defects arising out of faulty design or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

9. GUARANTEE TABLES:

- The tenderer shall fill-in schedules of the attached guarantee tables.
- By signing the tender, the tenderer guarantees that the equipment supplied will confirm to technical and performance data show in such tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.

GUARANTEE TABLES

M.V./L.V. VOLTAGE DETECTOR RANGE FROM 38.0 V TO 36K.V.



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No	Item	Guarantee	Supplier
1	Manufacturer's name		
2	Standard specifications applied	IEC 61243-1 IEC61235	
3	Voltage indication - Visual - Audible		
4	Nominal voltage range		
5	Nominal Frequency		
6	The threshold voltage - Response time - Time rating - Non response to DC voltage - Leakage current	Volt Sec Sec Yes $\leq 0.5\text{mA}$	
7	Environmental specifications - Operating temperature - Humidity - Storage temperature		
8	Universal link - Material - Length		
9	Insulated stick - Material - Length - Diameter - Length of handle - Height of hand guard - Grip force - Deflection - Dielectric strength	$\leq 200\text{n}$ ≤ 10 of voltage detector length 100k.v. r.m.s.:30cm apart:1sec	
10	Length of contact electrode extension		
11	Battery - Battery rating and type - Battery low voltage indication		

We guarantee the data given above for the equipment offered.

Signature:

Date: